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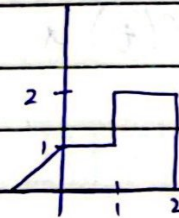
Roll No.:- 22L-7651

Section:- 4-A (EE)

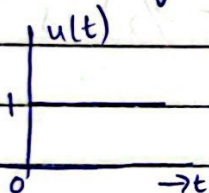
Course:- Signals & Systems (Assignment 1)

Question 1:-

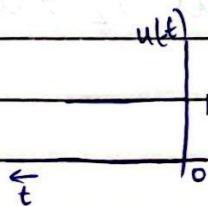
a) $x(t)u(1-t)$



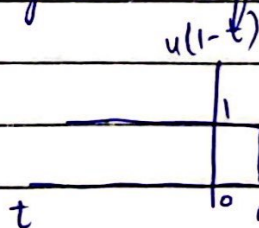
the step signal $u(t)$ is represented as



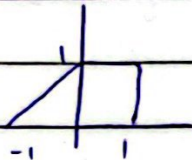
using time reversal, we need to find $u(-t)$ first



Now using time shift, we find $u(1-t)$

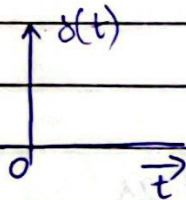


so the signal $u(1-t)$ exists from $-\infty$ to 1. So the combined signal $x(t)u(1-t)$ will also exist from $-\infty$ to 1. So, the signal $x(t)u(1-t)$ will be

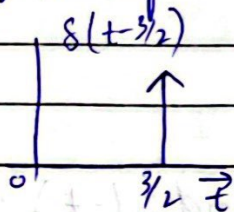


b) $x(t)\delta(t - 3/2)$

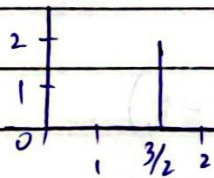
The graph of impulse $\delta(t)$ is



The graph of shifted impulse $\delta(t - 3/2)$ is



So, the signal $\delta(t - 3/2)$ exists at $3/2$ only. So the signal $x(t)\delta(t - 3/2)$ will be



Question 2:-

a) $\int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t-1) dt$

Here $f(t) = t^2 + \cos \pi t$

$$= f(t) \int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t-1) dt$$

$$= (1 + \cos \pi) \int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t-1) dt$$

$$= (1 - 1) \int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t-1) dt$$

$$= 0$$

b) $\int_{-\infty}^{\infty} (e^{-t}) \delta(2t-2) dt$

$$= \int_{-\infty}^{\infty} e^{-t} \delta[2(t-1)] dt$$

$$= \int_{-\infty}^{\infty} e^{-t} \frac{1}{|2|} \delta(t-1) dt$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} e^{-t} \delta(t-1) dt$$

$$= \frac{1}{2} e^{-t} \Big|_{t=1}$$

$$= \frac{1}{2e}$$

Question 3:-

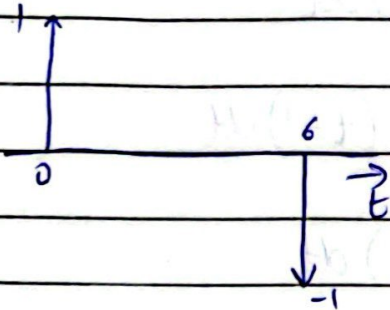
a) $x(t) = u(t) - u(t-6)$

$$\frac{dx(t)}{dt} = \frac{d}{dt} (u(t) - u(t-6))$$

$$\frac{dx(t)}{dt} = \delta(t) - \delta(t-6)$$

$$\begin{cases} \frac{d}{dt} u(t) = 2t \\ \frac{d}{dt} x(t) = u(t) \\ t(u(t)) = x(t) \end{cases}$$

sketch :-



$$b) x(t) = t [u(t) - u(t-6)]$$

$$= t u(t) - t u(t-6)$$

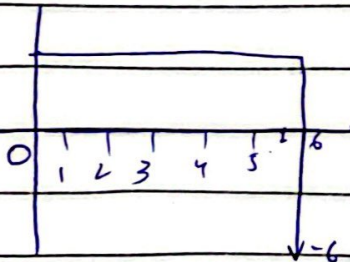
$$x(t) = t u(t) - (t-6) u(t-6) - 6 u(t-6)$$

$$= t u(t) - t u(t-6) - 6 u(t-6)$$

$$\frac{dx(t)}{dt} = \frac{d}{dt} [t u(t) - t u(t-6) - 6 u(t-6)]$$

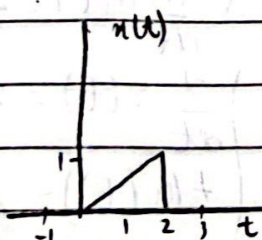
$$\frac{dx(t)}{dt} = u(t) - u(t-6) - 6 \delta(t-6)$$

sketch :-



Question 4 :-

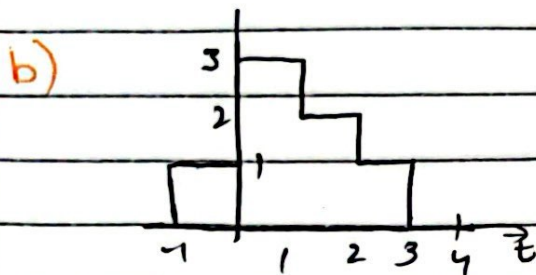
a)



the slope is $\frac{1}{2}$

$$x(t) = \frac{1}{2} r(t) - \frac{1}{2} \delta(t-2) - 4(t-2)$$

$$\left[x(t) = \frac{1}{2} t u(t) - \frac{1}{2} (t-2) u(t-2) - u(t-2) \right]$$



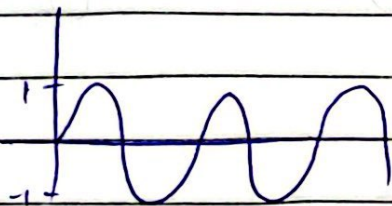
$$[x(t) = u(t+1) + 2u(t) - u(t-1) - 4(t-2) - u(t-3)]$$

$$(5-1)u(t) - (1-1)u(t-1) + (1-1)u(t-2)$$

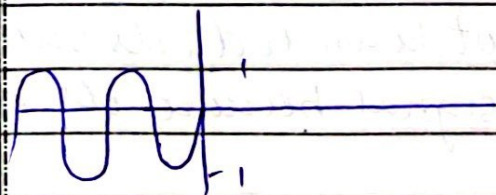
Question 5:-

Find and sketch even and odd component of $\sin \omega_0 t u(t)$

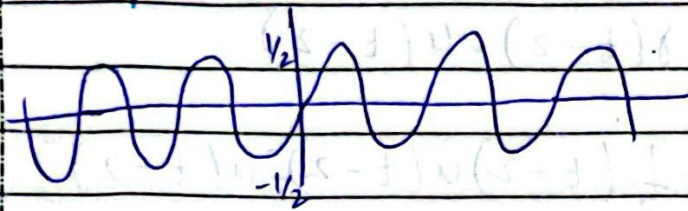
$\sin \omega_0 t u(t)$



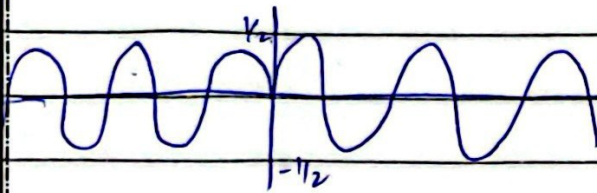
$\sin \omega_0 (-t) u(t)$



even part:-

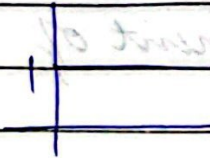


odd part:-

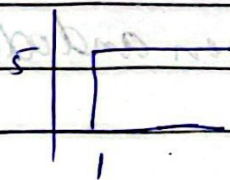


Question 6:-

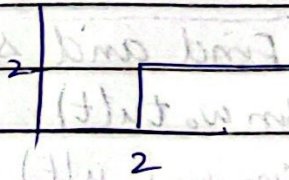
a) $u(t) + 5u(t-1) - 2u(t-2)$



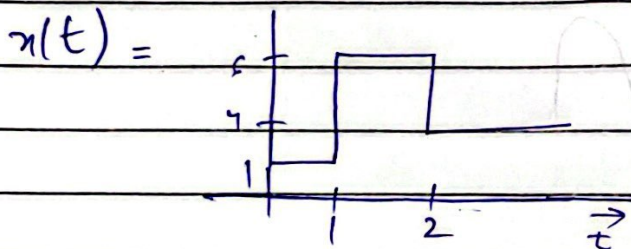
$u(t)$



$5u(t-1)$

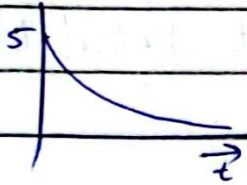


$2u(t-2)$



As we see this signal is not bounded, its value is not finite - So its power signal because its energy is infinite -

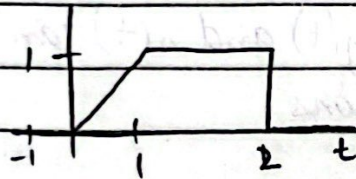
b) $(1 + e^{-5t})u(t)$



at $t \rightarrow \infty$ signal approaches towards zero. So $\int_{-\infty}^{\infty} [x(t)] < \infty$ infinite - So it's also energy signal.

Question 7:-

$y(t) = x(-2t-3)$



$x(t)$ can be described by its even and odd components.

$$x(t) = \underbrace{\frac{1}{2} [x(t) + x(-t)]}_{\text{even}} + \underbrace{\frac{1}{2} [x(t) - x(-t)]}_{\text{odd}}$$

$y(t)$ can be defined over 2 parts

• $t \in [0, 1];$

$$y(t) = t(u(t) - u(t-1))$$

• $t \in [1, 2];$

$$y(t) = u(t-1) - u(t-2)$$

concluding the expression of $y(t)$

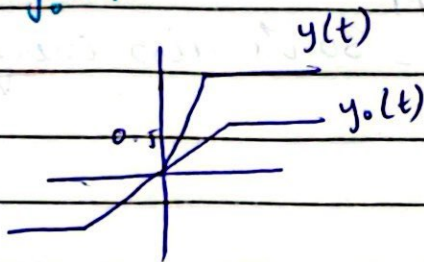
$$tu(t) + (1-t)u(t-1) - u(t-2)$$

computing the odd part of $y(t)$:

$$y_0(t) = \frac{1}{2} [y(t) - y(-t)]$$

$$= \frac{1}{2} [tu(t) + (1-t)u(t-1) - u(t-2) + tu(-t) - (1+t)u(-t-1) + 2u(t-2)]$$

sketch $y_0(t)$:



the relationship b/w $y(t)$ and $x(t)$ can be established by relevant signal operations.

1. Time shifting:-

$$\phi(t) = x(t-T)$$

2. Time expansion:-

$$\phi(t) = x\left(\frac{t}{a}\right)$$

Time compression:

$$\phi(t) = x(at)$$

3. Time reversal:-

$$\phi(t) = x(-t)$$

$$\text{given } y(t) = x(-2t-3)$$

Applying time shift by 3, the coordinates become $(t+3, y)$

Applying time compression by -2, the coordinates become $(\frac{t+3}{-2}, y)$

let t' be abscissa of $y(t)$, then

$$t' = \frac{t+3}{-2}$$

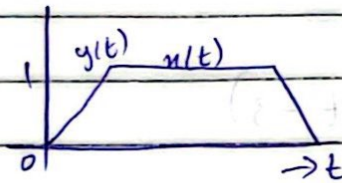
$$-2$$

$$t = -2t' - 3$$

So, $x(t)$ can be written as

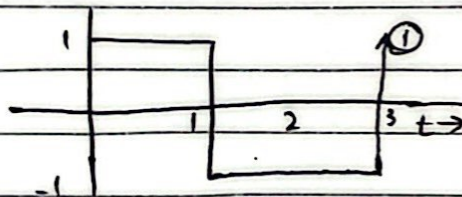
$$x(t) = y\left(-\frac{1}{2}t + 3\right)$$

sketch $x(t)$



Question 8:

$$\int_{-\infty}^t x(t)$$



Defining $x(t)$ in sections

- $\forall t \in [0, 1]:$

$$x_1(t) = u(t) - u(t-1)$$

- $\forall t \in [1, 3]:$

$$x_2(t) = -u(t-1) + u(t-3)$$

- $t = 3:$

$$x_3(t) = u(t) \delta(t-3) \Rightarrow u(3) \delta(t-3) = \delta(t-3)$$

concluding the expression of $x(t)$

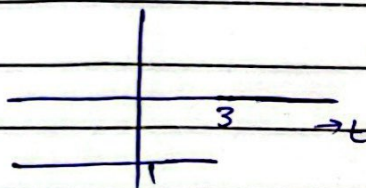
$$x(t) = x_1(t) + x_2(t) + x_3(t)$$

$$= u(t) - 2u(t-1) + u(t-3) + \delta(t-3)$$

the integral becomes

$$\begin{aligned} & \int_{-\infty}^t x(t) dt \\ &= \int_{-\infty}^t u(\tau) - 2u(\tau-1) + u(\tau-3) + \delta(\tau-3) d\tau \\ &= \int_0^t d\tau - 2 \int_1^t d\tau + \int_3^t d\tau + \int_{-\infty}^t \delta(\tau-3) d\tau \\ &= t - 2(t-1) + (t-3) + u(t-3) \\ &= -1 + u(t-3) \end{aligned}$$

sketch:



Question 9:

a) $\frac{dy}{dt} + 2y(t) = x^2(t)$

the response to $x_1(t)$ and $x_2(t)$ shall be

$$\frac{dy_1}{dt} + 2y_1(t) = x_1^2(t) \rightarrow \textcircled{1}$$

$$\frac{dy_2}{dt} + 2y_2(t) = x_2^2(t) \rightarrow \textcircled{2}$$

multiplying $\textcircled{1}$ by k_1 and $\textcircled{2}$ by k_2 and adding them

$$\frac{d(k_1 y_1 + k_2 y_2)}{dt} + 2(k_1 y_1 + k_2 y_2) = k_1 x_1^2(t) + k_2 x_2^2(t)$$

considering the input

$$x(t) = k_1 x_1(t) + k_2 x_2(t)$$

the equation become

$$\frac{d(k_1 y_1(t) + k_2 y_2(t))}{dt} + 2(k_1 x_1(t) + k_2 x_2(t)) = (k_1 x_1(t) + k_2 x_2(t))$$

As both input and output equations are not equal.
So, the system is not linear.

b) $\frac{dy}{dt} + 3ty(t) = t^2 x(t)$

the response to $x_1(t)$ and $x_2(t)$ shall be

$$\frac{dy_1}{dt} + 3ty_1(t) = t^2 x_1(t) \rightarrow \textcircled{1}$$

$$\frac{dy_2}{dt} + 3ty_2(t) = t^2 x_2(t) \rightarrow \textcircled{2}$$

multiplying $\textcircled{1}$ by k_1 and $\textcircled{2}$ by k_2 and adding them

$$\frac{d(k_1 y_1 + k_2 y_2)}{dt} + 3t(k_1 y_1(t) + k_2 y_2(t)) = t^2(k_1 x_1(t) + k_2 x_2(t))$$

considering the input

$$x(t) = k_1 x_1(t) + k_2 x_2(t)$$

the equation becomes

$$\frac{d(k_1 y_1 + k_2 y_2)}{dt} + 3t(k_1 y_1(t) + k_2 y_2(t)) = t^2(k_1 x_1(t) + k_2 x_2(t))$$

Hence the system is linear.

Question 10:-

$$y(t) = \sin(x(t)) \cos t$$

here

$$x(t) \rightarrow y(t)$$

$$x(t-t_0) \rightarrow y_1(t)$$

$$y(t-t_0) \rightarrow y_2(t)$$

$$x(t-t_0) = y_1(t) = \sin(\omega(t-t_0)) \cos t \rightarrow \textcircled{1}$$

$$y(t-t_0) = y_2(t) = \sin(\omega(t-t_0)) \cos(t-t_0) \rightarrow \textcircled{2}$$

As $\textcircled{1} \neq \textcircled{2}$

$$y_1(t) \neq y_2(t)$$

so the system is time variant system